The Oncologist
The official journal of the Society for Translational Oncology

Supplemental Tables for:
Towards a Treatment Sequencing Strategy: A Systematic Review of Treatment Regimens in Advanced Gastric Cancer/Gastroesophageal Junction Adenocarcinoma
Daniel Catenacci et al.

Supplemental Table 1. Key patient demographics and disease characteristics of the included 1L and post-1L RCTs ordered by publication year and therapy line

1L STUDY									
Trial Number (Acronym)	**Phase**	**Design (N)**	**Region/Center**	**Treatment**	**Patients, n**	**Male, n (%)**	**Median Age**	**ECOG PS, n (%)**	**Site of Primary Tumor, n (%)**
Kang, 2009 [24]	3	Open label (N = 316)	Multinational/ Multicenter Asian participation: Yes (66%)	Cis + Cap	160	103 (64%)	56 (range, 26-74)	NR	NR
				Cis + 5-FU	156	108 (69%)	56 (range, 22-73)	NR	NR
Curran, 2009 [29]	3	NR (N = 333)	NR Asian participation: NR	FOLFIRI	170	NR	NR	NR	NR
				Cis + FU	163	NR	NR	NR	NR
FLAGS Ajani, 2010 [11]	3	NR (N = 1,053)	US, Europe, South America, Australia/ Multicenter Asian participation: Yes (0.8%)	S-1 + Cis	527	382 (73.3%)	59.0 (range, 18-83)	▪ 0, 226 (43.3%) ▪ 1, 295 (56.6%)	▪ Stomach, 438 (84.1%) ▪ GEJ, 82 (15.7%) ▪ Stomach and GEJ, 1 (0.2%)

1L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
				FU + Cis	526	347 (68.3%)	60.0 (range, 20-85)	▪ 0, 200 (39.4%) ▪ 1, 308 (60.6%)	▪ Stomach, 417 (82.1%) ▪ GEJ, 88 (17.3%) ▪ Stomach and GEJ, 3 (0.6%)
ToGA Bang, 2010 [5]	3	Open label (N = 594)	Multinational/ Multicenter Asian participation: Yes (53%)	Tras + Chemo[a]	298	226 (77%)	Mean, 59.4 (SD, 10.8)	▪ 0-1, 264 (90%) ▪ 2, 30 (10%)	▪ Stomach, 236 (80%) ▪ GEJ, 58 (20%)
				Chemo[a]	296	218 (75%)	Mean, 58.5 (SD, 11.2)	▪ 0-1, 263 (91%) ▪ 2, 27 (9%)	▪ Stomach, 242 (83%) ▪ GEJ, 48 (17%)
AVAGAST Ohtsu, 2011 [33]	3	DBPC (N = 774)	Multinational/ Multicenter Asian participation: NR	Bev + Cis + Cap/FU[b]	387	257 (66%)	58 (range, 22-81)	▪ 0-1, 365 (94%) ▪ ≥2, 22 (6%)	▪ Stomach, 333 (86%) ▪ GEJ, 54 (14%)
				Placebo + Cis + Cap/FU[b]	387	258 (67%)	59 (range, 22-82)	▪ 0-1, 367 (95%) ▪ ≥2, 20 (5%)	▪ Stomach, 338 (87%) ▪ GEJ, 49 (13%)
EXPAND Lordick, 2013 [32]	3	Open label (N = 904)	Multinational/ Multicenter Asian participation: Yes (38%)	CTX + Cap + Cis	455	339 (75%)	60 (range, 51-66)	▪ 0, 237 (52%) ▪ 1, 218 (48%)	▪ Stomach, 376 (83%) ▪ GEJ, 71 (16%) ▪ Unknown, 8 (2%)

1L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
				Cap + Cis	449	334 (74%)	59 (range, 53-66)	▪ 0, 228 (51%) ▪ 1, 220 (49%)	▪ Stomach, 371 (83%) ▪ GEJ, 73 (16%) ▪ Unknown, 5 (1%)
REAL3 Waddell, 2013 [38]	3	Open label (N = 553)	UK/ Multicenter Asian participation: NR	Epir + Ox + Cap	275	226 (82%)	62 (IQR, 54-68)	WHO PS ▪ 0, 117 (43%) ▪ 1, 143 (52%) ▪ 2, 15 (5%)	▪ Esophagus, 111 (40%) ▪ GEJ, 75 (27%) ▪ Stomach, 89 (32%)
				Epir + Ox + Cap + Pan	278	232 (83%)	63 (IQR, 56-68)	WHO PS ▪ 0, 118 (42%) ▪ 1, 144 (52%) ▪ 2, 16 (6%)	▪ Esophagus, 106 (38%) ▪ GEJ, 94 (34%) ▪ Stomach, 78 (28%)
Van Cutsem, 2015 [2]	2	NR (N = 254)	US, Europe/ Multicenter Asian participation: NR	Doc + Ox	79	65%	Mean, 59	NR	▪ Stomach, 90%[b,c] ▪ GEJ, 37%[b,c]
				Doc + Ox + 5-FU/FOL	89	69%	Mean, 58	NR	▪ Stomach, 84%[b] ▪ GEJ, 39%[b]
				Doc +Ox + Cap	86	74%	Mean, 59	NR	▪ Stomach, 87% ▪ GEJ, 33%
Guimbaud, 2014 [36]	3	Open label (N = 416)	France/ Multicenter Asian participation: NR	Epir + Cis + Cap	209	154 (73.7%)	61.4 (range, 27.9-83.8)	NR	▪ Gastric, 133 (63.6%) ▪ GEJ, 73 (34.9%)

				FOLFIRI	207	155 (74.9%)	61.4 (range, 28.8-81.0)	NR	▪ Gastric, 138 (66.7%) ▪ GEJ, 63 (30.4%)
Kim, 2014 [31]	3	DBPC (N = 244)	South Korea/ Multicenter Asian participation: Yes	Sim + Cap + Cis	120	91 (75.8%)	53.5 (range, 20-78)	▪ 0-1, 119 (99.2%) ▪ 2, 1 (0.8%)	▪ Stomach, 94 (78.3%) ▪ GEJ, 6 (5.0%) ▪ Unknown, 20 (16.6%)
				Placebo + Cap + Cis	124	85 (68.5%)	54.5 (range, 24-79)	▪ 0-1, 120 (96.7%) ▪ 2, 4 (3.3%)	▪ Stomach, 99 (79.8%) ▪ GEJ, 6 (4.8%) ▪ Unknown, 19 (15.3%)
SOS Ryu, 2015 [1]	3	Open label (N = 615)	South Korea, Japan/ Multicenter Asian participation: Yes	S-1 (D 1-14) + Cis (D1)	306	231 (75%)	60 (range, 27-74)	▪ 0, 68 (22%) ▪ 1, 228 (75%) ▪ 2, 10 (3%)	NR
				S-1 (D 1-21) + Cis (D1 or 8)	309	233 (75%)	59 (range, 29-74)	▪ 0, 50 (16%) ▪ 1, 256 (83%) ▪ 2, 3 (1%)	NR
AVATAR Shen, 2015 [30]	3	DBPC (N = 202)	China/ Multicenter Asian participation: Yes	Placebo + Cap + Cis	102	74 (72.5%)	Mean, 55.5	▪ 0-1, 97 (95.1%) ▪ ≥ 2, 5 (4.9%)	▪ Stomach, 82 (80.4%) ▪ GEJ, 20 (19.6%)
				Bev + Cap + Cis	100	68 (68%)	Mean, 54.2	▪ 0-1, 95 (95.0%) ▪ ≥ 2, 5 (5.0%)	▪ Stomach, 85 (85.0%) ▪ GEJ, 15 (15.0%)
JapicCTI-101021 Yamada, 2015 [22]	3	Open label (N = 685)	Japan/ Multicenter Asian participation: Yes	S-1 + Ox	343	240 (75.5%)	65 (range, 21-83)	▪ 0, 224 (70.4%) ▪ 1, 91 (28.6%) ▪ 2, 3 (0.9%)	NR

				S-1 + Cis	342	237 (73.1%)	65 (range, 29-85)	▪ 0, 228 (70.4%) ▪ 1, 92 (28.4%) ▪ 2, 4 (1.2%)	NR
Li, 2015 [67]	Pilot	Open label (N = 236)	China/ Single center Asian participation: Yes	S-1 + Cis	120	84 (70%)	53.3 (12.1) [mean, SD]	▪ 0, 28 (23.30%) ▪ 1, 85 (70.83%) ▪ 2, 7 (5.83%)	NR
				5-FU + Cis	116	85 (73.3%)	55.3 (11.2) [mean, SD]	▪ 0, 29 (25.00%) ▪ 1, 83 (71.55%) ▪ 2, 4 (3.45%)	NR
Wang, 2016 (35)	3	Open label (N = 243)	China/ Multicenter Asian participation: Yes	Doc + Cis + FU	121	81 (68.1%)	56.6 (range, 19-80)	NR	▪ GEJ, 20 (16.8%) ▪ Fundus, 12 (10.1%) ▪ Antrum, 44 (37%) ▪ Body, 43 (36.1%) ▪ Other, 13 (10.9%) ▪ Unknown, 1 (0.8%)
				Cis + FU	122	88 (76.5%)	55.5 (range, 33-74)	NR	▪ GEJ, 29 (25.2%) ▪ Fundus, 9 (7.8%) ▪ Antrum, 41 (35.7%) ▪ Body, 37 (32.2%) ▪ Other, 6 (5.2%) ▪ Unknown, 2 (1.7%)
G-SOX Bando, 2016 [23]	3	Open label (N = 685)	NR Asian participation: NR	S-1 + Ox (≥70 years)	116	91 (78.4%)	73	▪ 0, 86 (74.1%) ▪ 1, 27 (23.3%) ▪ 2, 3 (2.6%)	NR
				S-1 + Cis (≥70 years)	104	81 (77.9%)	73	▪ 0, 66 (63.5%) ▪ 1, 37 (35.6%) ▪ 2, 1 (1.0%)	NR

				S-1 + Ox (< 70 years)	227	168 (74.0%)	62	▪ 0, 155 (68.3%) ▪ 1, 71 (31.3%) ▪ 2, 1 (0.4%)	NR
				S-1 + Cis (< 70 years)	238	169 (71.0%)	62	▪ 0, 176 (73.9%) ▪ 1, 59 (24.8%) ▪ 2, 3 (1.3%)	NR
TRIO-013/ LOGIC Hecht, 2016 [4]	3	DBPC (N = 545)	Multinational/ Multicenter Asian participation: NR	Lap + Cap + Ox	272	189 (76%)	61 (range, 19-86)	▪ 0, 79 (32%) ▪ 1, 149 (60%) ▪ 2, 21 (8%)	▪ Esophageal, 12 (5%) ▪ Gastric, 214 (86%) ▪ GEJ, 23 (9%)
				Placebo + Cap + Ox	273	176 (74%)	59 (range, 27-84)	▪ 0, 63 (26%) ▪ 1, 153 (64%) ▪ 2, 22 (9%)	▪ Esophageal, 8 (3%) ▪ Gastric, 210 (88%) ▪ GEJ, 20 (8%)
DIGEST Ajani, 2017 [25]	3	NR (N = 361)	Multinational/ Multicenter North America/Western Europe/Eastern Europe/Rest of World	S-1 + Cis	239	124 (51.9%)	56 (range, 25-86)	▪ 0, 74 (31.0%) ▪ 1, 165 (69.0%)	▪ Stomach, 223 (93.3%) ▪ GEJ, 16 (6.7%)
				5-FU + Cis	122	60 (49.2%)	56 (range, 27-83)	▪ 0, 38 (31.1%) ▪ 1, 83 (68.0%)	▪ Stomach, 117 (95.9%) ▪ GEJ, 5 (4.1%)
RILOMET-1 Catenacci, 2017 [8]	3	Double-blind (N = 609)	Multinational/ Multicenter 27 countries (Europe, North and South America, and Australia)	Rilotumumab	304	205 (67%)	61 (range, 28-84)	▪ 0, 117 (39%) ▪ 1, 187 (62%) ▪ 2, 0	▪ Distal esophageal, 24 (8%) ▪ GEJ, 53 (17%) ▪ Gastric, 227 (75%)

				Placebo	305	220 (72%)	59 (range, 26-81)	▪ 0, 155 (38%) ▪ 1, 189 (62%) ▪ 2, 1 (< 1%)	▪ Distal esophageal, 39 (13%) ▪ GEJ, 71 (23%) ▪ Gastric, 195 (64%)
METGastric Shah, 2017 [9]	3	Double-blind (N = 562)	Multinational/ Multicenter	Onartuzumab	279	188 (67.4%)	NR	▪ 0, 112 (40.4%) ▪ ≥ 1, 162 (58.1%) ▪ Not done, 5 (1.8%)	▪ Stomach, 214 (76.7%) ▪ GEJ, 65 (23.3%)
				Placebo	283	183 (64.7%)	NR	▪ 0, 118 (41.7%) ▪ ≥ 1, 158 (55.8%) ▪ Not done, 7 (2.5%)	▪ Stomach, 218 (77.0%) ▪ GEJ, 65 (23.0%)
Shu, 2017 (28)	NR	Open label (N = 349)	China/ Single center	Ox + tegafur	164	93 (56.7%)	NR	NR	▪ Cardiac, 37 (22.6%) ▪ Gastric body, 79 (48.2%)
				Ox + S-1	168	88 (52.4%)	NR	NR	▪ Cardiac, 34 (20.2%) ▪ Gastric body, 88 (52.4%)
HELOISE Shah, 2017 [6]	3b	Open label (N = 248)		SoC Tras	124	95 (76.6%)	60.0 (range, 26-83)	▪ 0 or 1, 0 ▪ 2, 124 (100%)	▪ Stomach, 95 (76.6%) ▪ GEJ, 29 (23.4%)
				Higherdose Tras	124	95 (76.6%)	65.0 (range, 27-82)	▪ 0 or 1, 2 (1.6%) ▪ 2, 122 (98.4%)	▪ Stomach, 101 (81.5%) ▪ GEJ, 23 (18.5%)
Lu, 2018 [26]	3	Open label (N = 320)	China/ multicenter	Pac + Cap	160	115 (71.9%)	Mean (SD): 56.6 (11.3)	NR	▪ GEJ, 68 (42.5%)
				Cis + Cap	160	118 (73.8%)	Mean (SD): 56.2 (10.9)	NR	▪ GEJ, 63 (39.4%)

Lu, 2019 [27]	NR	Open label (N = 300)	China/ Multicenter	S-1 + Doc	150	NR	NR	NR	NR
				S-1 + Cis	150	NR	NR	NR	NR
RAINFALL Fuchs, 2019 [34)	3	Double-blind (N = 645)	Multinational/ Multicenter (126 centers in 20 countries)	Ram + Cis + Cap/5-FU[c]	326	214 (66%)	60 (IQR, 51-68)	▪ 0, 141 (43%) ▪ 1, 185 (57%)	▪ Gastric, 242 (74%) ▪ GEJ, 83 (25%) ▪ Data missing, 1 (<1%)
				Placebo + Cis + Cap/5-FU[c]	319	215 (67%)	60 (IQR, 54-68)	▪ 0, 143 (45%) ▪ 1, 176 (55%)	▪ Gastric, 239 (75%) ▪ GEJ, 80 (25%) ▪ Data missing, 0
JACOB Tabernero, 2018 [7]	3		Multinational/ Multicenter (197 clinics in 30 countries)	Pertuzumab + Tras/chemo	388	294 (76%)	62.0 (IQR, 54.5-69)	▪ 0, 162 (42%) ▪ 1, 226 (58%)	▪ GEJ, 110 (28%) ▪ Stomach, 278 (72%)
				Placebo + Tras/chemo	392	323 (82%)	61.0 (IQR, 54-68)	▪ 0, 162 (41%) ▪ 1, 229 (59%)	▪ GEJ, 98 (25%) ▪ Stomach, 294 (75%)
JCOG1013 Yamada, 2019 [37]	3		Japan	Doc + Cis/S-1	370	257 (69%)	65 (IQR, 58–70)	▪ 0, 239 (66%) ▪ 1, 131 (35%)	NR
				Cis/S-1	371	256 (69%)	65 (IQR, 59–70)	▪ 0, 241 (65%) ▪ 1, 130 (35%)	NR

2L STUDY

Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
Thuss-Patience, 2011 [15]	3	Open label (N = 40)	Germany/ Multicenter Asian participation:	Iri + BSC	21	18 (86%)	58 (range, 43-73)	▪ 0-1, 17 (81%) ▪ 2, 4 (19%)	▪ GEJ, 9 (43%) ▪ Gastric body, 12 (57%)

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
			NR						
				BSC	19	11 (58%)	55 (range, 35-72)	▪ 0-1, 14 (74%) ▪ 5 (26%)	▪ GEJ, 8 (42%) ▪ Gastric body, 11 (58%)
Yi, 2012 [64]	2	Open label (N = 105)	South Korea/ Single-center Asian participation: Yes	Doc + sunitinib	56	40 (71.4%)	54 (range, 20-72)	▪ 0, 2 (3.6%) ▪ 1, 28 (87.8%) ▪ 2, 6 (10.7%)	NR
				Doc	49	33 (67.3%)	52 (range, 36-70)	▪ 0, 3 (6.1%) ▪ 1, 43 (85.7%) ▪ 2, 3 (6.1%)	NR
WJOG 4007 Hironaka, 2016 [40]	3	Open label (N = 219)	Japan/ Multicenter Asian participation: Yes	Pac	108	84 (77.7%)	64.5 (range, 37-75)	▪ 0-1, 104 (96.3%) ▪ 2, 4 (3.7%)	▪ Gastric, 106 (97%) ▪ GEJ, 3 (3%)
				Iri	111	87 (78.4%)	65 (range, 38-75)	▪ 0-1, 107 (96.4%) ▪ 2, 4 (3.6%)	▪ Gastric, 107 (94%) ▪ GEJ, 7 (6%)
Sym, 2013 [49]	2	Open label (N = 59)	South Korea/ single center Asian participation: Yes	Iri	29	20 (69%)	60 (range, 45-76)	▪ 0-1, 27 (93%) ▪ 2, 2 (7%)	▪ Stomach, 27 (93%) ▪ GEJ, 2 (7%)
				FOLFIRI[a]	30	14 (47%)	61 (range, 30-75)	▪ 0-1, 27 (90%) ▪ 2, 3 (10%)	▪ Stomach, 27 (90%) ▪ GEJ, 3 (10%)
Roy, 2013 [12]	2	NR (N = 132)	UK, Spain, Taiwan,	PEP02[d]	44	35 (79.5%)	56 (range, 38-81)	▪ 0-1, 41 (93%) ▪ 2, 3 (7%)	▪ Gastric, 37 (84%) ▪ GEJ, 7 (16%)

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
			Croatia, South Korea, Bosnia/ Multinational Asian participation: Yes						
				Iri	44	34 (77.3%)	62 (range, 33-79)	▪ 0-1, 41 (93%) ▪ 2, 3 (7%)	▪ Gastric, 35 (80%) ▪ GEJ, 9 (20%)
				Doc	44	34 (77.3%)	58 (range, 33-81)	▪ 0-1, 40 (91%) ▪ 2, 4 (9%)	▪ Gastric, 30 (68%) ▪ GEJ, 14 (32%)
TyTAN Satoh, 2014 [63]	3	Open label (N = 261)	China, Japan, South Korea, Taiwan/ Multicenter Asian participation: Yes	Lap + Pac	132	101 (77%)	61.0 (range, 32-79)	▪ 0, 60 (45%) ▪ 1, 72 (55%)	▪ Gastric, 132 (100%) ▪ Other, 0 (0%)
				Pac	129	106 (82%)	62.0 (range, 22-80)	▪ 0, 48 (37%) ▪ 1, 81 (63%)	▪ Gastric, 127 (98%) ▪ Other, 2 (2%)
COUGAR-02 Ford, 2014 [52]	3	Open label (N = 168)	UK/ Multicenter Asian participation: NR	Doc	84	69 (82%)	65 (range, 28-84)	▪ 0, 24 (28%) ▪ 1, 46 (55%) ▪ 2, 14 (17%)	▪ Esophagus, 18 (22%) ▪ GEJ, 27 (32%) ▪ Stomach, 39 (46%)
				Active symptom control	84	67 (80%)	66 (range, 36-84)	▪ 0, 22 (26%) ▪ 1, 50 (60%) ▪ 2, 12 (14%)	▪ Esophagus, 15 (18%) ▪ GEJ, 32 (38%) ▪ Stomach, 37 (44%)
TCOG GI-0801/	3	Open label	Japan/	Iri + Cis	64	49	66 (range, 29-	▪ 0, 44 (69%)	NR

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
BIRIP Higuchi, 2014 [54]		(N = 130)	Multicenter Asian participation: Yes			(77%)	80)	▪ 1, 20 (31%) ▪ 2, 0	
				Iri	63	55 (87%)	67 (range, 49-78)	▪ 0, 43 (68%) ▪ 1, 20 (32%) ▪ 2, 0	NR
REGARD Fuchs, 2014 [39]	3	DBPC (N = 355)	North, Central and America; Europe; Asia; Australia; Africa/ Multicenter Asian participation: Yes (16%)	Ramucirumab	238	169 (71%)	60 (IQR, 52-67)	▪ 0, 67 (28%) ▪ 1, 171 (72%) ▪ 2, 0	▪ Gastric, 178 (75%) ▪ GEJ, 60 (25%)
				Placebo	117	79 (68%)	60 (IQR, 51-71)	▪ 0, 31 (26%) ▪ 1, 85 (73%) ▪ 2, 1 (1%)	▪ Gastric, 87 (74%) ▪ GEJ, 30 (26%)
RAINBOW Wilke, 2014 [66]	3	DBPC (N = 665)	North and South America; Europe; Asia; Australia/ Multicenter Asian participation: Yes (35%)	Ram + Pac	330	229 (69%)	61 (range, 25-83)	▪ 0, 117 (35%) ▪ 1, 213 (65%)	▪ Gastric, 264 (80%) ▪ GEJ, 66 (20%)
				Placebo + Pac	335	243 (73%)	61 (range, 24-84)	▪ 0, 144 (43%) ▪ 1, 191 (57%)	▪ Gastric, 106 (97%) ▪ GEJ, 3 (3%)

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
Kim, 2015 [55]	2	NR (N = 52)	South Korea/ Multicenter Asian participation: Yes	Doc	27	24 (89%)	54	▪ 0, 5 (18.5%) ▪ 1, 21 (77.8%) ▪ 2, 1 (3.7%)	NR
				Doc + Ox	25	18 (72%)	59	▪ 0, 3 (12.0%) ▪ 1, 21 (84.0%) ▪ 2, 1 (4.0%)	NR
JapicCTI-090849 Satoh, 2015 [62]	2	Open label (N = 83)	Japan, South Korea/ Multicenter Asian participation: Yes	Nimotuzumab + Iri	40	33 (82.5%)	60 (range, 27-75)	▪ 0, 19 (47.5%) ▪ 1, 21 (52.5%)	▪ GEJ, 4 (18.2%) ▪ Gastric region, 18 (81.8%)
				Iri	42	33 (78.6%)	63.5 (range, 32-75)	▪ 0, 17 (40.5%) ▪ 1, 25 (59.5%)	▪ GEJ, 1 (3.8%) ▪ Gastric region, 25 (96.2%)
JACCRO GC05 Tanabe, 2015 [53]	2/3	Open label (N = 304)	Japan/ Multicenter Asian participation: Yes	S-1 + Iri	145	99 (68%)	67 (range, 37-84)	▪ 0, 117 (81%) ▪ 1, 28 (19%)	NR
				Iri	148	109 (74%)	66 (range, 22-83)	▪ 0, 116 (78%) ▪ 1, 32 (22%)	NR

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
TRICS Nishikawa, 2015 [57]	3	Open label (N = 168)	Japan/ Multicenter Asian participation: Yes	Iri + Cis	84	68 (81%)	67 (range, 36-85)	▪ 0, 68 (81%) ▪ 1, 16 (19%)	NR
				Iri	84	63 (75%)	68 (range, 35-87)	▪ 0, 68 (81%) ▪ 1, 16 (19%)	NR
Study 39 Bang, 2015 [65]	2	DBPC (N = 124)	South Korea/ Multicenter Asian participation: Yes	Olaparib + Pac	62	49 (79%)	63.0 (range, 31-77)	▪ 0, 32 (51.6%) ▪ 1, 30 (48.4%) ▪ 2, 0	NR
				Placebo + Pac	62	44 (71%)	60.5 (range, 25-79)	▪ 0, 28 (45.2%) ▪ 1, 32 (51.6%) ▪ 2, 2 (3.2%)	NR
RAINBOW (subgroup analysis [East Asia]) Muro, 2016 [68]	3	DBPC (N = 223)	Subgroup analysis, East Asian patients Asian participation: Yes	Ram + Pac	109	73 (67%)	62 (range, 28-76)	▪ 0, 44 (40%) ▪ 1, 65 (60%)	▪ Gastric, 107 (94%) ▪ GEJ, 7 (6%)
				Placebo + Pac	114	81 (71%)	62 (range, 28-81)	▪ 0, 60 (53%) ▪ 1, 54 (47%)	▪ Gastric, 65 (95.6%) ▪ GEJ, 3 (4.4%)

2L STUDY									
Trial Number (Acronym)	**Phase**	**Design (N)**	**Region/Center**	**Treatment**	**Patients, n**	**Male, n (%)**	**Median Age**	**ECOG PS, n (%)**	**Site of Primary Tumor, n (%)**
RAINBOW (subgroup analysis [Japan]) Shitara, 2016 [69]	3	DBPC (Japanese subgroup N = 140; Western subgroup, N = 398)	Japanese subgroup Asian participation: Yes	Ram + Pac	68	46 (67.6%)	64.0 (range, 34-76)	▪ 0, 37 (54.4%) ▪ 1, 31 (45.6%)	▪ Gastric, 106 (97%) ▪ GEJ, 3 (3%)
				Placebo + Pac	72	53 (73.6)	64.5 (range, 29-76)	▪ 0, 43 (59.7%) ▪ 1, 29 (40.3%)	▪ Gastric, 106 (97%) ▪ GEJ, 3 (3%)
			Western subgroup Asian participation: NR	Ram + Pac	198	140 (70.7%)	60.0 (range, 25-83)	▪ 0, 68 (34.3%) ▪ 1, 130 (65.7%)	▪ Gastric, 107 (94%) ▪ GEJ, 7 (6%)
				Placebo + Pac	200	147 (73.5%)	61.0 (range, 24-84)	▪ 0, 75 (37.5%) ▪ 1, 125 (62.5%)	▪ Gastric, 65 (95.6%) ▪ GEJ, 3 (4.4%)
CCOG0701 Nakanishi, 2016 [56]	2	NR (N = 81)	Japan/ Multicenter Asian participation: Yes	Pac	40	34 (85%)	62 (range, 38-80)	▪ 0, 31 (78%) ▪ 1, 6 (15%) ▪ 2, 3 (7%)	NR
				Pac + S-1	49	29 (76%)	64 (range, 42-79)	▪ 0, 30 (79%) ▪ 1, 7 (18%) ▪ 2, 1 (3%)	NR

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
JCOG0407 Nishina, 2016 [41]	2	Open-label (N = 100)	Japan/ Multicenter Asian participation: Yes	Best available 5-FU	49	33 (67%)	59 (range, 30-74)	▪ 0, 25 (51%) ▪ 1, 23 (47%) ▪ 2, 1 (2%)	NR
				Pac	51	36 (71%)	64 (range, 39-75)	▪ 0, 27 (53%) ▪ 1, 22 (43%) ▪ 2, 2 (4%)	NR
Bang, 2017a [42]	2	Open label (N = 114)	Asia, Europe, and North America Asian participation: Yes	Ipilimumab	57	36 (63.2%)	65 (range, 34-86)	▪ 0, 32 (56.1%) ▪ 1, 25 (43.9%)	▪ Gastric, 48 (84.2%) ▪ GEJ, 9 (15.8%)
				BSC	57	41 (71.9%)	62 (range, 32-80)	▪ 0, 24 (42.1%) ▪ 1, 33 (57.9%)	▪ Gastric, 47 (82.5%) ▪ GEJ, 10 (17.5%)
GOLD Bang, 2017b [59]	3	Double-blind, placebo-controlled (N = 525)	Asia (China, Japan, South Korea, Taiwan) Asian participation: Yes	Olaparib + Pac	263	174 (66%)	58 (range, 49-67)	▪ 0, 117 (44%) ▪ 1, 145 (55%) ▪ Missing, 1 (<1%)	NR
				Placebo + Pac	262	185 (71%)	59 (range, 50-65)	▪ 0, 107 (41%) ▪ 1, 154 (59%) ▪ Missing, 1 (<1%)	NR
				[Subgroup]: ATM-negative tumors[e]: Olaparib + Pac	48	31 (65%)	58 (range, 50-66)	▪ 0, 22 (46%) ▪ 1, 26 (54%) ▪ Missing, 0	NR

2L STUDY									
Trial Number (Acronym)	**Phase**	**Design (N)**	**Region/Center**	**Treatment**	**Patients, n**	**Male, n (%)**	**Median Age**	**ECOG PS, n (%)**	**Site of Primary Tumor, n (%)**
				[Subgroup]: ATM-negative tumors[e]: Placebo + Pac	46	31 (67%)	60·5 (range, 57-69)	▪ 0, 23 (50%) ▪ 1, 23 (50%) ▪ Missing, 0	NR
GATSBY Thuss-Patience, 2017 [44]	2/3	Open label (N = 345)[f]	Multinational (28 countries) Asian participation: Yes	Taxane	117	95 (81%)	62.0 (range, 27-80)	▪ 0, 43 (37%) ▪ 1, 73 (62%) ▪ 2, 1 (1%) ▪ 3, 0	▪ Stomach, 84 (72%) ▪ GEJ, 33 (28%)
				Trastuzumab emtansine	228	177 (78%)	62.0 (range, 19-79)	▪ 0, 99 (43%) ▪ 1, 128 (56%) ▪ 2, 0 ▪ 3, 1 (<1%)	▪ Stomach, 151 (66%) ▪ GEJ, 77 (34%)
SHINE Van Cutsem, 2017 [45]	2	Open label (N = 71)	Asia, North America, Europe Asian participation: Yes	AZD4547	41	29 (70.7%)	60.6 (SD, 11.4)	NR	NR
				Pac	30	22 (73.3%)	61.9 (SD, 10.7)	NR	NR
Lee, 2017 [13]	2	Open label (N = 72)	South Korea (3 centers) Asian participation: Yes	Doc	23	18 (78.3%)	56 (range, 34-68)	▪ 0, 2 (8.7%) ▪ 1, 21 (91.3%) ▪ 2, 0	NR
				Doc + Cis	23	20 (87.0%)	55 (range, 38-74)	▪ 0, 1 (4.3%) ▪ 1, 20 (87.0%) ▪ 2, 2 (8.7%)	NR
				Doc + S-1	23	14	55 (range, 39-	▪ 0, 1 (4.3%)	NR

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
						(60.9%)	68)	▪ 1, 20 (87.0%) ▪ 2, 2 (8.7%)	
ABSOLUTE Shitara, 2017 [14]	3	Open label (N = 741)	Japan (72 centers) Asian participation: Yes	Nab-Pac Q3W	243	178 (73%)	66 (range, 60-72)	▪ 0, 167 (69%) ▪ 1, 72 (30%) ▪ 2, 4 (2%)	NR
				Nab-Pac QW	240	178 (74%)	67 (range, 60-72)	▪ 0, 168 (70%) ▪ 1, 70 (29%) ▪ 2, 2 (1%)	NR
				Pac	243	176 (72%)	65 (range, 59-71)	▪ 0, 168 (69%) ▪ 1, 71 (29%) ▪ 2, 4 (2%)	NR
DREAM Kang, 2018 [48]	3	Open label (N = 238)	South Korea/ Multicenter Asian participation: Yes	DHP107 (oral Pac)	118	91 (77.1%)	59 (range, 33-83)	▪ 0/1, 116 (98.3%)	NR
				IV Pac	118	94 (79.7%)	59 (range, 27-80)	▪ 0/1, 117 (99.2%)	NR
KEYNOTE-061 Shitara, 2018 [43]	3	Open label (N = 592)	Multinational (30 countries) Asian participation: Yes	Pembrolizumab	296	202 (68%)	62.5 (range, 54-70)	▪ 0, 127 (43%) ▪ 1, 169 (57%) ▪ 2, 0	▪ Stomach, 207 (70%) ▪ GEJ, 89 (30%)
				Pac	296	208 (70%)	60.0 (53-68)	▪ 0, 137 (46%) ▪ 1, 158 (53%) ▪ 2, 1 (<1%)	▪ Stomach, 200 (68%) ▪ GEJ, 96 (32%)
				[Subgroup]:	196	146	64.0 (57-70.5)	▪ 0, 88 (45%)	▪ Stomach, 134

2L STUDY									
Trial Number (Acronym)	**Phase**	**Design (N)**	**Region/Center**	**Treatment**	**Patients, n**	**Male, n (%)**	**Median Age**	**ECOG PS, n (%)**	**Site of Primary Tumor, n (%)**
				PD-L1 CPS ≥1[f]: Pembrolizumab		(74%)		▪ 1, 108 (55%) ▪ 2, 0	(68%) ▪ GEJ, 62 (32%)
				[Subgroup]: PD-L1 CPS ≥1[f]: Pac	199	140 (70%)	61.0 (54-68)	▪ 0, 92 (46%) ▪ 1, 106 (53%) ▪ 2, 1 (< 1%)	▪ Stomach, 126 (63%) ▪ GEJ, 73 (37%)
KCSG ST10-01 Lee, 2019 [46]	3	Open label (N = 112)	South Korea (16 centers) Asian participation: Yes	Pac	54	38 (70.4%)	58.5 (range, 38-82)	▪ 1, 52 (96.3%) ▪ 2, 2 (3.7%)	▪ Stomach, 51 (94.4%) ▪ GEJ, 3 (5.6%)
				Iri	58	40 (69.0%)	59 (range, 38-77)	▪ 1, 56 (96.6%) ▪ 2, 2 (3.4%)	▪ Stomach, 51 (87.9%) ▪ GEJ, 7 (12.1%)
Combined second- and third-line population									
Kang, 2012 (51)	3	Open label (N = 202)	South Korea (8 centers) Asian participation: Yes	Doc or Iri + BSC	133	93 (70%)	56 (range, 31-83)	▪ 0, 72 (54%) ▪ 1, 61 (46%)	NR
				BSC	69	44 (64%)	56 (range, 32-74)	▪ 0, 36 (52%) ▪ 1, 33 (48%)	NR
GRANITE-1 Ohtsu, 2013 [60]	3	Double-blind (N = 656)	Multinational Asian participation: Yes	Everolimus + BSC	439	322 (73%)	62 (range, 20-86)	▪ 0, 144 (33%) ▪ 1, 269 (61%) ▪ 2, 25 (6%) ▪ Missing, 1 (< 1%)	▪ Proximal stomach, 162 (37%) ▪ Distal stomach, 276 (63%) ▪ Missing, 1 (<1%)
				Placebo + BSC	217	161 (74%)	62 (range, 26-88)	▪ 0, 70 (32%) ▪ 1, 120 (55%) ▪ 2, 27 (12%)	▪ Proximal stomach, 94 (43%) ▪ Distal stomach, 123 (57%)

2L STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
								▪ Missing, 0 (0%)	▪ Missing, 0 (0%)
Shitara, 2014 [47]	2	Open label (N = 89)	Japan (13 centers) Asian participation: Yes	Dose-escalated Pac	44	32 (73%)	62 (range, 29-78)	▪ 0, 22 (50%) ▪ 1, 18 (41%) ▪ 2, 4 (9%)	NR
				Pac	45	29 (64%)	65 (range, 33-80)	▪ 0, 23 (51%) ▪ 1, 18 (40%) ▪ 2, 4 (9%)	NR
Fushida, 2016 [58]	2	Open label (N = 66)	Japan (18 centers) Asian participation: Yes	Pac	33	NR	68 (range, 49-84)	▪ 0-1, 28 (85%) ▪ 2, 5 (15%)	NR
				Pac + valproic acid	33	NR	67 (range, 31-83)	▪ 0-1, 29 (88%) ▪ 2, 4 (12%)	NR
Moehler, 2016 [50]	2	Double-blind, placebo-controlled (N = 90)	Germany (15 centers) Asian participation: No	Na-FOLFIRI + sunitinib	45	33 (73%)	62 (range, 37-76)	NR	▪ Stomach, 22 (49%) ▪ Cardia, 23 (51%) ▪ Unknown, 0 (0%)
				Na-FOLFIRI + placebo	45	30 (67%)	57 (range, 28-84)	NR	▪ Stomach, 23 (51%) ▪ Cardia, 20 (44%) ▪ Unknown, 2 (4%)

2L STUDY

Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
INTEGRATE Pavlakis, 2016 [61]	2	Double-blind (N = 147)	53 centers in four countries: Australia, Canada, New Zealand, and South Korea Asian participation: Yes	Regorafenib + BSC	97	78 (80%)	63 (range, 33-81)	▪ 0, 41 (42%) ▪ 1, 56 (58%)	▪ Stomach or other, 60 (62%) ▪ GEJ, 37 (38%)
				Placebo + BSC	50	40 (80%)	62 (range, 32-85)	▪ 0, 21 (42%) ▪ 1, 29 (58%)	▪ Stomach or other, 31 (62%) ▪ GEJ, 19 (38%)

3L+ STUDY

Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
Li, 2013 [21]	2	DBPC parallel-arm (N = 144)	China	Apatinib 850 mg once daily	47	39 (83%)	55	▪ 0, 3 (6%) ▪ 1, 44 (94%)	▪ Liver, 28 (60%) ▪ Lungs, 5 (11%) ▪ Posterior peritoneum lymph node, 9 (19%)
				Apatinib 425 mg twice daily	46	34 (74%)	53	▪ 0, 2 (4%) ▪ 1, 44 (96%)	▪ Liver, 20 (43%) ▪ Lungs, 11 (24%) ▪ Posterior peritoneum lymph node, 11 (24%)
				Placebo	48	36 (75%)	54	▪ 0, 1 (2%) ▪ 1, 47 (98%)	▪ Liver, 23 (48%) ▪ Lungs, 9 (19%) ▪ Posterior

3L+ STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
									peritoneum lymph node, 11 (23%)
Li, 2016 [19]	3	DBPC (N = 267)	China	Apatinib 850 mg once daily	176	132 (75%)	58 (range, 23-71)	▪ 0, 48, (27.3%) ▪ 1, 128 (72.7%)	▪ Gastric, 69 (68.3%) ▪ GEJ, 22 (21.8%) ▪ Unknown, 10 (9.9%)
				Placebo	91	69 (75.8%)	58 (range, 28-70)	▪ 0, 15 (16.5%) ▪ 1, 76, (83.5%)	▪ Gastric, 43 (71.7%) ▪ GEJ, 14 (23.3%) ▪ Unknown, 3 (5%)
ATTRACTION-2 Kang, 2017 [17]	3	DBPC (N = 493)	Japan South Korea, and Taiwan/Multicenter Asian participation: Yes	Nivolumab	330	229 (69%)	62 (range, 54-69)	▪ 0, 95 (29%) ▪ 1, 235 (71%)	▪ Gastric, 272 (82.4%) ▪ Esophagogastric junction, 30 (9.1%) ▪ Unknown, 28 (8.5%)
				Placebo	163	119 (73%)	61 (range 53-68)	▪ 0, 48 (29%) ▪ 1, 115 (71%)	▪ Gastric, 135 (82.8%) ▪ Esophagogastric junction, 12 (7.4%) ▪ Unknown, 16 (9.8%)
JAVELIN Gastric 300 Bang, 2018 [16]	3	Open-label (N = 371)	Australia, Belgium, Czechia, France, Germany, Italy, Poland, South Korea, Spain, and US/Multicenter Asian participation: Yes	Avelumab + BSC	185	140 (75.7)	59 (range, 29-86)	▪ 0, 66 (35.7%) ▪ 1, 119 (64.3%)	▪ Gastric, 122 (65.9%) ▪ GEJ, 63 (34.1%)
				Chemotherapy + BSC	186	127 (68.3)	61 (range, 18-82)	▪ 0, 62 (33.3%) ▪ 1, 124 (66.7%)	▪ Gastric, 138 (74.2%)

3L+ STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
									▪ GEJ, 48 (25.8%)
TAGS Shitara, 2018 [18]	3	DBPC (N = 507)	Belarus, Belgium, Czech Republic, France, Germany, Ireland, Israel, Italy, Japan, Poland, Portugal, Romania, Russia, Spain, Turkey, the UK, and the US/Multinational Asian participation: Yes	Trifluridine/tipiracil + BSC	337	252 (75%)	64 (IQR, 56-70)	▪ 0, 123 (36%) ▪ 1, 214 (64%)	▪ Gastric, 239 (71%) ▪ GEJ, 98 (29%) ▪ Both, 0
				Placebo + BSC	170	117 (69%)	63 (IQR, 56-69)	▪ 0, 68 (40%) ▪ 1, 102 (60%)	▪ Gastric, 121 (71%) ▪ GEJ, 47 (28%) ▪ Both, 2 (1%)
CheckMate-032 Janjigian, 2018 [20]	1/2	Open-label, (N = 160)	Canada, Denmark, Finland, Germany, Italy, Spain, UK and US/Multicenter Asian participation: Yes	Nivolumab 3 mg/kg	59	45 (76%)	60 (range, 29-80	▪ 0, 29 (49%) ▪ 1, 30 (51%)	▪ Gastric, 19 (32%) ▪ Esophageal, 9 (15%) ▪ GEJ, 31 (53%)
				Nivolumab 1 mg/kg + ipilimumab 3 mg/kg	49	34 (69%)	53 (range, 27-77)	▪ 0, 27 (55%) ▪ 1, 22 (45%)	▪ Gastric, 22 (45%) ▪ Esophageal, 8 (16%) ▪ GEJ, 19 (39%)
				Nivolumab 3 mg/kg + ipilimumab 1 mg/kg	52	45 (87%)	58 (range, 19-81)	▪ 0, 18 (35%) ▪ 1, 34 (65%)	▪ Gastric, 18 (35%) ▪ Esophageal, 9 (17%) ▪ GEJ, 25 (48%)

3L+ STUDY Trial Number (Acronym)	Phase	Design (N)	Region/Center	Treatment	Patients, n	Male, n (%)	Median Age	ECOG PS, n (%)	Site of Primary Tumor, n (%)
ATTRACTION-2 (subgroup analysis [Japanese patients]) Kato, 2019 [70]	3	DBPC (N = 226)	See Kang, 2017 [17] (primary study)	Nivolumab	152	111 (73%)	65 (range, 20-83)	▪ 0, 64 (42%) ▪ 1, 88 (58%)	NR
				Placebo	74	57 (77%)	66 (range, 28-79)	▪ 0, 31 (42%) ▪ 1, 43 (58%)	NR
ATTRACTION-2 (subgroup analysis [Japanese patients]) Satoh, 2019 [10]	3	DBPC (N = 507)	See Kang, 2017 [17] (primary study)	History of trastuzumab nivolumab	59	47 (79.7%)	62 (range, 23-83)	▪ 0, 19 (32.2%) ▪ 1, 40 (67.8%)	NR
				History of trastuzumab placebo	22	20 (90.9%)	62.5 (range, 33-77)	▪ 0, 9 (40.9%) ▪ 1, 13 (59.1%)	NR
				No history of trastuzumab nivolumab	271	182 (67.2%)	62 (range, 20-83)	▪ 0, 69 (25.5%) ▪ 1, 202 (74.5%)	NR
				No history of trastuzumab placebo	141	99 (70.2%)	61 (range, 26-83)	▪ 0, 38 (27.0%) ▪ 1, 103 (73.0%)	NR

ATM, ataxia-telangiectasia mutated (a key activator of DNA damage response; Bev, bevacizumab; BSC, best supportive care; Cap, capecitabine; Cis, cisplatin; CPS, combined positive score; CTX, cetuximab; D, day; Doc, docetaxel; DBPC, double-blind placebo-controlled; ECOG, Eastern Cooperative Oncology Group; Epir, epirubicin; FOLFIRI, irinotecan + 5-fluorouracil + folinic acid (also known as leucovorin); FOL, folinic acid; FP, fluoropyrimidine (5-FU or capecitabine); FU, fluorouracil; GEJ, gastroesophageal junction; IQR, interquartile range; Iri, irinotecan; IV, intravenous; Lap, lapatinib; NR, not reported; Ox, oxaliplatin; Pan, panitumumab; PD-L1, programmed death-ligand 1; PS, performance status; Q3W, every 3 weeks; QW, weekly; Ram, ramucirumab; SD, standard deviation; Sim, simvastatin; SoC, standard of care; Tras, trastuzumab; UK, United Kingdom; US, United States.

Supplemental Table 2. Included first-line and post-first-line studies reporting significant time to progression, time to treatment failure, objective response rate, duration of response, and/or disease control rate results

Clinical Endpoint	No. of 1L Studies Reporting	No. of 1L studies with significant findings	Studies		Treatment Arms		Median, mo or Rate, %		*P*-value
					Arm 1	Arm 2	Arm 1	Arm 2	
TTP	5	1	**ToGA** Bang, 2010 [5]		Tras + CF	CF	7.1	5.6	0.0003
TTF	5	4	**FLAGS** Ajani, 2010 [11]		S+Cis	F+Cis	3.8	3.8	0.032
			Wang, 2016 [35]		mDCF	CF	3.4	2.4	0.0027
			G-SOX Bando, 2016 [23]	≥70 yrs	S+Ox	S+Cis	5.5	4.3	0.008
				<70 yrs	S+Ox	S+Cis	4.2	4.2	0.234
			Guimbaud,2014 [36]		FOLFIRI > ECCp	ECCp > FOLFIRI	5.1	4.2	0.008
ORR	20	9	**ToGA** Bang, 2010 [5]		Tras + CF	CF	47%	35%	0.0017
			TRIO-013/LOGIC Hecht, 2016 [4]		Lap + CpO	CpO	53%	39%	0.0031
			Wang, 2016 [35]		mDCF	CF	48.7%	33.9%	0.0244
			Kang, 2009 [24]		CCp	CF	46%	32%	0.020
			AVAGAST Ohtsu, 2011 [33]		Bev+CCp/F	CCp/F	46%	37.4%	0.0315
			DIGEST Ajani, 2017 [25]		S-1 + Cis	5-FU + Cis	34.7%	19.8%	0.0122
			RILOMET-1 Catenacci, 2017 [8]		Rilotumumab + Epir + Cis + Cap	Placebo + Epir + Cis + Cap	29.8%	44.6%	0.0005

			Lu, 2018 [26]	Pac + Cap	Cis + Cap	43.1%	28.8%	0.01
DoR	6	2	**ToGA** Bang, 2010 [5]	Tras + CF	CF	6.9	4.8	<0.0001
			RAINFALL[v]	Ram + Cis + Cap/5-FU[w]	Placebo + Cis + Cap/5-FU[w]	5.7	4.3	0.0026
			Fuchs, 2019 [34]					
DCR	7	1	**RILOMET-1** Catenacci, 2017 [8]	Rilotumumab + Epir + Cis + Cap	Placebo + Epir + Cis + Cap	53.4%	70.8%	< 0.0001

Clinical Endpoint	**No. of 2L or combined 2L and 3L Studies Reporting**	**No. of 2L or combined 2L and 3L studies with significant findings**	**Studies**	**Treatment Arms**		**Rate, %**		***P*-value**
				Arm 1	**Arm 2**	**Arm 1**	**Arm 2**	
TTP	5	0						
TTF	4	0						
ORR	26	3	**TyTAN** Satoh, 2014 [63]	Lap + Pac	Pac	27%	9%	0.001
			Yi, 2012 [64]	Doc + Sun	Doc	41.1%	14.3%	0.002
			RAINBOW Wilke 2014 [66]	Ram + Pac	Pac	28%	16%	0.0001
			RAINBOW Shitara 2016 [69] Japanese	Ram + Pac	Pac	41.2%	19.4%	0.0035
			Western	Ram + Pac	Pac	26.8%	13%	0.0004
			RAINBOW Muro, 2016 [68] East Asian	Ram + Pac	Pac	34%	20%	0.0134
DoR	6	0						
DCR	16	5	**Yi**, 2012 [65]	Doc + Sun	Doc	75%	51%	0.011

			TCOG GI-0801/BIRIP Higuchi, 2014 [54]	IC	Iri	75%	54%	0.0162
			REGARD Fuchs, 2014 [39]	Ram	Placebo	49%	23%	<0.0001
			RAINBOW Wilke, 2014 [66]	Ram + Pac	Pac	80%	64%	<0.0001
			RAINBOW Shitara, 2016 [69] Japanese	Ram + Pac	Pac	94.1%	75%	0.0021
			Western	Ram + Pac	Pac	76.8%	56.5%	<0.0001
			RAINBOW Muro, 2016 [68] East Asian	Ram + Pac	Pac	89%	75%	0.0058
			Kim, 2015 [55]	Doc + Ox	Doc	76%	48.1%	0.02
Clinical Endpoint	**No. of 3L and Beyond Studies Reporting**	**No. of 3L and Beyond studies with significant findings**	**Studies**	**Treatment Arms**		**Rate, %**		**P-value**
				Arm 1	**Arm 2**	**Arm 1**	**Arm 2**	
ORR	**8**	**2**	**ATTRACTION-2 (subgroup analysis [Japanese patients])** Kato, 2019 [70]	Nivolumab	Placebo	14.0	0	P = 0.0023
			Li, 2103 [21]	Apatinib 850 mg once daily	Placebo	51.1	10.42	P <.001
				Apatinib 425 mg twice daily	Placebo	34.8	10.42	P <.001
DCR	**8**	**3**	**Li,** 2016 [19]	Apatinib	Placebo	42.05	8.79	P < .001
			ATTRACTION-2 (subgroup analysis	Nivolumab	Placebo	45.0	23.3	P = 0.0037

	[Japanese patients]) Kato, 2019 [70]					
	TAGS Shitara, 2018 [18]	Trifluridine/tipiracil + BSC	Placebo + BSC	44	14	P < 0·0001

Bev, bevacizumab; BSC, best supportive care; Cis, cisplatin; DCR, disease control rate**;** Doc, docetaxel; DoR, duration of response; Epir, epirubicin; F, fluorouracil; Iri, irinotecan; Lap, lapatinib; FOLFIRI, folinic acid+fluorouracil+irinotecan; mDCF, modified DCF; NR, not reported;ORR, objective response rate; Ox, oxaliplatin; Pac, paclitaxel; Ram, ramucirumab; S, S-1; Sun, sunitinib; TTP, time to progression; TTF, time to treatment failure; Tras, trastuzumab;

Supplemental Table 3. Overview of adverse events and toxicities reported

1L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
SOS Ryu, 2015 [1]	S-1 days 1-14 + cisplatin day 1 (n = 306)	(98%)	One treatment-related death	NR	One treatment-related death	One treatment-related death	NR	220/330 (73%) (grade 3 or 4)	
	S-1 days 1-21 + cisplatin on day 1 or 8 (n = 309)	(97%)	One treatment-related death	NR	One treatment-related death	One treatment-related death	NR	153/303 (51%) (grade 3 or 4)	
AVATAR Shen, 2015 [30]	Placebo + capecitabine + cisplatin (n = 102)	Any grades 3-5 69/101 (68.3%)	21/101 (20.8%)	NR	(8.0%)	NR	NR	NR	NR
	Bevacizumab + capecitabine + cisplatin (n = 100)	Any grades 3-5 60/100 (60.0%)	15/100 (15.0%)	NR	(4.0%)	NR	NR	NR	NR
REAL3 Waddell, 2013 [38]	Epirubicin + oxaliplatin + capecitabine (n = 275)	NR	NR	NR	NR	6/266 (2.3%)	NR	NR	NR
	Epirubicin + oxaliplatin + capecitabine + panitumumab (n = 278)	NR	NR	NR	NR	4/276 (1.4%)	NR	NR	NR
JapicCTI-101021 Yamada, 2015 [22]	S-1 + oxaliplatin (n = 318)	NR	17/328 (5.2%)	NR	NR	4/338 (1.2%)	99/338 (29.3%)	NR	NR
	S-1 + cisplatin (N = 324)	NR	30/332 (9.0%)	NR	NR	8/335 (2.4%)	127/335 (37.9%)	NR	NR

1L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
G-SOX Bando, 2016 [23]	S-1 + oxaliplatin (≥70 years) (n = 116)	NR	5/114 (4.4%)	NR	NR	NR	NR	NR	NR
	S-1 + cisplatin (≥70 years) (n = 104)	NR	14/101 (13.9%)	NR	NR	NR	NR	NR	NR
	S-1 + oxaliplatin (<70 years) (n = 227)	NR	5/224 (2.2%)	NR	NR	NR	NR	NR	NR
	S-1 + cisplatin (<70 years) (n = 238)	NR	14/234 (6.0%)	NR	NR	NR	NR	NR	NR
FLAGS Ajani, 2010 [11]	S-1 + cisplatin (n = 521)	NR	34/521 (6.5%)	NR	1/527 (0.2%)	13/521 2.5%	NR	NR	NR
	Fluorouracil + cisplatin (n = 508)	NR	55/508 (10.8%)	NR	2/5268 (0.4%)	25/508 4.9%	NR	NR	NR
ToGA Bang, 2010 [5]	Trastuzumab plus chemotherapy (n = 294)	292 (99%)	NR	246 (84%)	NR	10 (3%)	201 (68%)	201 (68%)	201 (68%)
	Chemotherapy alone (n = 290)	284 (98%)	NR	237 (82%)	NR	3 (1%)	198 (68%)	198 (68%)	198 (68%)
Curran, 2009 [29]	Irinotecan, folinic acid, and 5-fluorouracil (FOLFIRI) (n = 170)	NR	10.0%	NR	NR	1 (0.6%)	NR	NR	NR
	Cisplatin and 5-fluorouracil (n = 163)	NR	21.5%	NR	NR	5 (3.1%)	NR	NR	NR
Guimbaud, 2014 [36]	Epirubicin, cisplatin, and capecitabine (n = 209)	NR	Toxicity, 14.5%	NR	9 (6.5%)	NR	NR	84%	84%
	Fluorouracil, leucovorin, and irinotecan (n = 207)	NR	Toxicity, 3.9%	NR	7 (10.0%)	NR	NR	69%	69%
TRIO-013/ LOGIC Hecht, 2016 [4]	Lapatinib + capecitabine + oxaliplatin (n = 249)	255 (94%)	57 (21%)	NR	NR	15 (6%)	72 (27%)	NR	NR
	Placebo + capecitabine + oxaliplatin (n = 238)	236 (88%)	50 (19%)	NR	NR	9 (3%)	52 (19%)	NR	NR
Kang, 2009 [24]	Cisplatin + oral capecitabine (n = 156)	NR	28/156 (18%)	NR	NR	NR	NR	NR	20/156 (13%)
	Cisplatin + 5-FU (n = 155)	NR	28/155 (18%)	NR	NR	NR	NR	NR	16/155 (10%)

1L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
Kim, 2014 [31]	Simvastatin + capecitabine-cisplatin (n = 120)	NR	NR	NR	NR	NR	NR	63/120 (52.5%)[e]	
	Placebo + capecitabine-cisplatin (n = 124)	NR	NR	NR	NR	NR	NR	70/124 (56.4%)[e]	
Li, 2015 [67]	S-1 + cisplatin (n = 120)	NR	NR	NR	NR	NR	NR	76/120 (60%)	
	5-FU + cisplatin (n = 116)	NR	NR	NR	NR	NR	NR	47/116 (36%)	
EXPAND Lordick, 2013 [32]	Capecitabine-cisplatin + cetuximab (n = 446)	NR	72/446 (16%)	NR	NR	NR	NR	NR	NR
	Capecitabine-cisplatin (n = 436)	NR	80/436 (18%)	NR	NR	NR	NR	NR	NR
AVAGAST Ohtsu, 2011 [33]	Bevacizumab + cisplatin + capecitabine/fluorouracil (n = 386)	NR	81/386 (21%)[b]	NR	NR	7/386 (2%)	NR	(66%)[f]	
	Placebo + cisplatin + capecitabine/fluorouracil (n = 381)	NR	71/381 (19%)[b]	NR	NR	12/381 (3%)	NR	(64%)[f]	
Van Cutsem, 2015 [2]	Docetaxel + oxaliplatin (TE) (n = 79)	NR	NR	NR	4 (5%)	NR	NR	NR	
	Docetaxel + oxaliplatin + 5-FU (TEF) (n = 89)	NR	NR	NR	3 (3%)	NR	NR	NR	
	Docetaxel + oxaliplatin + capecitabine (TEX) (n = 86)	NR	NR	NR	11 (13%)	NR	NR	NR	
DIGEST Ajani, 2017 [25]	S-1 + cisplatin (n = 230)	214 (93.0%)	NR	NR	NR	1 (<1%)	63 (27.4%)	157 (68.3%)	NR
	5-FU + cisplatin (n = 118)	111 (94.1%)	NR	NR	NR	1 (<1%)	31 (26.3%)	78 (66.1%)	NR
RILOMET-1 Catenacci, 2017 [8]	Rilotumumab + Epir + Cis + Cap (n = 298)	283 (95%)	41 (14%)	3 (1%)	42 (14%)	NR	142 (48%)	NR	
	PBO + Epir + Cis + Cap (n = 299)	288 (96%)	30 (10%)	0	31 (10%)	NR	149 (50%)	NR	

1L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
METGastric Shah, 2017 [9]	Onartuzumab + mFOLFOX6 (n = 279)	274 (98.2%)	87 (31.2%)	NR	70 (25.1%)	NR	100 (35.8%)	192 (68.8)	
	PBO + mFOLFOX6 (n = 280)	273 (97.5%)	61 (21.8%)	NR	73 (26.1%)	NR	91 (32.5%)	187 (66.8)	
HELOISE Shah, 2017 [6]	SoC trastuzumab (n = 124)	113/124 (91.1%)	3 (2.4%)	NR	58 (46.8%)	NR	30 (24.2%)	74 (59.7%)	
	Higher dose trastuzumab (n = 124)	112/123 (91.1%)	4 (3.3%)	NR	67 (54.5%)	NR	35 (28.5%)	75 (61.0%)	
Lu, 2018 [26]	Paclitaxel + capecitabine (n = 158)	132 (83.5%)	NR	NR	NR	0	6 (3.8%)	54 (34.2%)	
	Cisplatin + capecitabine (n = 147)	130 (88.4%)	NR	NR	NR	1 (0.7%)	4 (2.7%)	59 (40.1%)	
Lu, 2019 [27]	S-1 + docetaxel (n = 150)	85.33%	1 (0.7%)	NR	NR	NR	NR	26.00%	
	S-1 + cisplatin (n = 150)	87.33%	2 (1.3%)	NR	NR	NR	NR	24.67%	
RAINFALL Fuchs, 2019 [34]	Ramucirumab + cisplatin + capecitabine/5-FU	NR	54/323 (17%)	16/323 (5%)	18/323 (6%)	7/323 (2.2%)	160/323 (50%)	NR	
	Placebo + cisplatin + capecitabine/5-FU	NR	39/315 (12%)	9/315 (3%)	19/315 (6%)	7/315 (2.2%)	149/315 (47%)	NR	
JACOB Tabernero, 2018 [7]	Pertuzumab + trastuzumab/chemotherapy	381/385 (99%)	45/385 (12%)	NR	240/385 (62.3%)	27/385 (7%)	48/385 (12%)	NR	
	Placebo + trastuzumab/chemotherapy	385/388 (99%)	45/388 (12%)	NR	262/388 (67.5%)	30/388 (8%)	40/388 (10%)	NR	
JCOG1013 Yamada, 2019 [37]	Docetaxel + cisplatin/S-1	NR	NR	40/370 (11%)	NR	NR	NR	NR	
	Cisplatin/S-1	NR	NR	39/371 (11%)	NR	NR	NR	NR	

2L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
JapicCTI-090849 Satoh, 2015 [62]	Nimotuzumab + irinotecan (n = 40)	NR	6/40 (15%)	NR	NR	2/40 (5.0%)	NR	NR	NR
	Irinotecan alone (n = 42)	NR	7/42 (16.7%)	NR	NR	0 (0%)	NR	NR	NR
TyTAN Satoh, 2014 [63]	Lapatinib + paclitaxel (n = 132)	131/131 (100%)	21/132 (16%)	NR	6/132 (4.6%)	4/132 (3.0%)	(26%)	60/132 (46%)	
	Paclitaxel alone (n = 129)	127/129 (98%)	12/129 (9%)	NR	4/129 (3.1%)	1/129 (0.2%)	(15%)	58/129 (45%)	
Sym, 2013 [49]	Irinotecan (n = 29)	NR	4/29 (14%)	NR	1/29 (3%)	1/29 (3.5%)	NR	NR	NR
	Irinotecan + leucovorin/5-FU (FOLFIRI) (n = 30)	NR	3/30 (10%)	NR	0/30 (0%)	0/30 (0%)	NR	NR	NR
JACCRO GC-05 Tanabe, 2015 [53]	S-1 + irinotecan (n = 145)	NR	NR	NR	NR	0/145 (0%)	NR	NR	NR
	Irinotecan (n = 148)	NR	NR	NR	NR	2/149 (1.3%)	NR	NR	NR
Thuss-Patience, 2011 [15]	Irinotecan + BSC (n = 21)	NR	NR	NR	All patients died	NR	NR	NR	NR
	BSC (n = 19)	NR	NR	NR	NR	NR	NR	NR	NR
Yi, 2012 [64]	Docetaxel + sunitinib (n = 56)	NR	NR	29/56 (51.8%)	NR	NR	NR	NR	NR
	Docetaxel alone (n = 49)	NR	NR	17/49 (34.7%)	NR	NR	NR	NR	NR

2L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
Study 39 Bang, 2015 [65]	Olaparib/paclitaxel (n = 62)	NR	1	25 (41.0%)	0	0	17 (27.9%)	46 (75.4%)	
	Placebo/paclitaxel (n = 62)	NR	5	10 (16.1%)	0	0	23 (37.1%)	46 (74.2%)	
COUGAR-02 Ford, 2014 [52]	Docetaxel (n = 84)	NR	20/84 (23.8%)	NR	7/84 (8.3%)	NR	NR	NR	17 (21%)
	Active symptom control (n = 84)	NR	NR	NR	32/84 (38.1%)	NR	NR	NR	3 (4%)
TCOG GI-0801/ BIRIP Higuchi, 2014 [54]	Irinotecan 60 mg/m^2 plus cisplatin 30 mg/m^2 (n = 64)	NR	3/64 (4.7%)	NR	NR	0/64 (0%)	NR	NR	NR
	Irinotecan alone (n = 63)	NR	3/66 (4.5%)	NR	NR	1/66 (1.5%)	NR	NR	NR
REGARD Fuchs, 2014 [39]	Ramucirumab (n = 236)	223/236 (94%)	25/236 (11%)	NR	20/236 (9%)	5/236 (2%)	NR	134/236 (57%)[f]	
	Placebo (n = 115)	101/115 (88%)	7/115 (6%)	NR	13/115 (3%)	2/115 (2%)	NR	67/115 (58%)[f]	
RAINBOW Wilke, 2014 [66]	Ramucirumab + paclitaxel (n = 327)	324/327 (99%)	39/327 (12%)	Ram, 16/327 (5%) Pac, 78/327 (24%)	12/327 (4%)	6/327 (2%)[b]	153/327 (47%)	155/327 (47%)	72/327 (22%)
	Placebo + paclitaxel (n = 329)	322/329 (98%)	38/329 (11%)	Pla, 3/329 (< 1%) Pac, 24/329 (7%)	13/329 (4%)	5/329 (2%)[b]	139/329 (42%)	128/329 (39%)	27/329 (8%)
RAINBOW (subgroup analysis) Muro, 2016 [68]	Ramucirumab + paclitaxel (n = 108)	NR	NR	NR	NR	NR	NR	NR	NR
	Placebo + paclitaxel (n = 112)	NR	NR	NR	NR	NR	NR	NR	NR

2L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
RAINBOW (subgroup analysis) Shitara, 2016 [69]	Japanese Ramucirumab + paclitaxel (n = 68)	68/68 (100%)	(7.4%)	NR	NR	NR	NR	57/68 (83.8%)[c]	57/68 (83.8%)[c]
	Japanese Placebo + paclitaxel (n = 71)	70/71 (98.6%)	(8.3%)	NR	NR	NR	NR	37/71 (52.1%)[c]	37/71 (52.1%)[c]
	Western Ramucirumab + paclitaxel (n = 196)	193/196 (98.5%)	(13.6%)	NR	NR	NR	NR	155/196 (79.1%)[c]	155/196 (79.1%)[c]
	Western Placebo + paclitaxel (n = 197)	191/197 (97%)	(13.5%)	NR	NR	NR	NR	122/197 (61.9%)[c]	122/197 (61.9%)[c]
WJOG 4007 Hironaka, 2013 [40]	Paclitaxel (n = 108)	NR	NR	NR	NR	0/108 (0%)	NR	NR	NR
	Irinotecan (n = 111)	NR	NR	NR	NR	2/111 (1.8%)	NR	NR	NR
Kim, 2015 [55]	Docetaxel (n = 27)	NR	NR	(3.7%)	NR	NR	NR	NR	NR
	Docetaxel + oxaliplatin (n = 25)	NR	NR	(2.2%)	NR	NR	NR	NR	NR
CCOG0701 Nakanishi, 2016 [57]	Paclitaxel (n = 41)	NR	4/41 (10%)	NR	NR	NR	NR	(25%)	(25%)
	S-1 + paclitaxel (n = 38)	NR	7/40 (19.4%)	NR	NR	NR	NR	(26%)	(26%)
JCOG0407 Nishina, 2016 [56]	Best available 5-FU (n = 49)	NR	(10.2%)	NR	NR	1/49 (2%)	(6.1%)	NR	NR
	Weekly paclitaxel (n = 51)	NR	(15.7%)	NR	NR	0/51 (0%)	(2%)	NR	NR
Roy, 2013 [12]	PEP02 (n = 44)	NR	6/44 (13.6%)	NR	NR	NR	NR	(38.6%)[g]	
	Irinotecan (n = 44)	NR	6/44 (13.6%)	NR	NR	NR	NR	(34.1%)[g]	
	Docetaxel (n = 44)	NR	6/44 (13.6%)	NR	NR	NR	NR	(15.9%)[g]	
Bang, 2017a [42]	Ipilimumab (n = 57)	41/57 (71.9%)[b]	10/57 (17.5%)[b]	NR	1/57 (1.8%)	0	17/57 (29.8%)[b]	13/57 (22.8%)[e]	

2L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
	BSC (n = 45)	25/45 (55.6%)[b]	4/45 (8.9%)[b]	NR	0	0	0/45 (0%)[b]	4/45 (8.9%)[e]	
GOLD Bang, 2017b [59]	Olaparib + paclitaxel (n = 262)	260/262 (99%)	41/262 (16%)	51/262 (19%)	166/262 (63%)	1/262 (<1%)	92/262 (35%)	NR	NR
	Placebo + paclitaxel (n = 259)	254/259 (98%)	25/259 (10%)	27/259 (10%)	193/259 (75%)	1/259 (<1%)	65/259 (25%)	NR	NR
GATSBY Thuss-Patience, 2017 [44]	Taxane (n = 111)	108/111 (97%)	15/111 (14%)	NR	69/111 (62%)	NR	31/111 (28%)	NR	NR
	Trastuzumab emtansine (n = 224)	218/224 (97%)	31/224 (14%)	NR	162/224 (72%)	NR	65/224 (29%)	NR	NR
SHINE Van Cutsem, 2017 [45]	AZD4547 (n = 40)	29/40 (72.5%)	3/40 (7.5%)	NR	27/40 (71.1%)	NR	1/40 (2.5%)	NR	NR
	Paclitaxel (n = 27)	19/27 (70.4%)	3/27 (11.1%)	NR	18/27 (60.0%)	NR	1/27 (3.7%)	NR	NR
Lee, 2017 [13]	Docetaxel (n = 23)	NR	1/23 (4.3%)	NR	NR	0	NR	NR	NR
	Docetaxel + cisplatin (n = 24)	NR	0	NR	NR	1/24 (4.2%)[c]	NR	NR	NR
	Docetaxel + S-1 (n = 23)	NR	1/23 (4.3%)	NR	NR	1/23 (4.3%)[c]	NR	NR	NR
ABSOLUTE Shitara, 2017 [14]	Nab-paclitaxel every 3 weeks (n = 244)	244/244 (100%)	41/244 (17%)	NR	NR	1/244 (0.4%)	NR	NR	NR
	Nab-paclitaxel (n = 241)	238/241 (99%)	17/241 (7%)	NR	NR	2/241 (0.8%)	NR	NR	NR
	Paclitaxel (n = 243)	243/243 (100%)	12/243 (5%)	NR	NR	1/243 (0.4%)	NR	NR	NR
DREAM Kang, 2018 [48]	DHP107 (oral paclitaxel) (n = 118)	115/118 (97.5%)	9/118 (7.6%)	NR	1/118 (0.8%)	0	46/118 (39%)	81/118 (68.6%)	
	IV paclitaxel (n = 118)	116/118 (98.7%)	16/118 (13.6%)	NR	0	0	49/118 (41%)	99/118 (83.9%)	

2L STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of AEs, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
KEYNOTE-061 Shitara, 2018 [43]	Pembrolizumab (n = 294)	155/294 (53%)	9/294 (3%)	NR	NR	3/294 (1%)	NR	NR	NR
	Paclitaxel (n = 276)	232/276 (84%)	15/276 (5%)	NR	NR	1/276 (< 1%)	NR	NR	NR
KCSG ST10-01 Lee, 2019 [46]	Paclitaxel (n = 52)	105/109 (96.3%)	NR	NR	1/52 (1.9%)	NR	7/52 (13.5%)	NR	NR
	Irinotecan (n = 57)		NR	NR	4/57 (7.0%)	NR	13/57 (22.8%)	NR	NR
Combined second- and third-line population									
GRANITE-1 Ohtsu, 2013 [60]	Everolimus + BSC (n = 439)	433/437 (99.1%)	94/437 (21.5%)	242/437 (55.4%)	NR	3/437 (0.7%)	NR	310/437 (71%)	
	Placebo + BSC (n = 217)	208/215 (96.7%)	34/215 (15.8%)	46/215 (21.4%)	NR	2/215 (0.9%)	NR	114/215 (53%)	
Shitara, 2014 [47]	Dose-escalated paclitaxel (n = 44)	NR	10/44 (22.7%)	NR	NR	0 (0%)	NR	NR	NR
	Paclitaxel (n = 45)	NR	8/45 (18%)	NR	NR	0 (0%)	NR	NR	NR
Fushida, 2016 [58]	Paclitaxel (n = 33)	5/33 (15%)	NR	NR	NR	NR	NR	5/33 (15 %)	
	Paclitaxel + valproic acid (n = 31)	1/31 (3.2%)	NR	NR	NR	NR	NR	1/31 (3.2%)	
Moehler, 2016 [50]	Na-FOLFIRI + sunitinib (n = 45)	43 (95.5%)	1 (2.2%)	NR	NR	NR	NR	33 (73.3%)	
	Na-FOLFIRI + placebo (n = 45)	42 (93.3%)	3 (6.7%)	NR	NR	NR	NR	31 (68.9%)	
INTEGRATE Pavlakis, 2016 [61]	Regorafenib + BSC (n = 97)	NR	14/100 (14%)	NR	5/100 (5%)	NR	31/97 (32%)	65/97 (67%)	
	Placebo + BSC (n = 50)	NR	0/52 (0%)	NR	3/52 (5.8%)	NR	9/50 (18%)	26/50 (52%)	

3L+ STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of Aes, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
Li, 2013 [21]	Apatinib (850 mg once daily) (n =47)	NR	NR	12.8%	1 (%)	NR	NR	NR	
	Apatinib (425 mg twice daily) (n = 46)	NR	NR	32.6%	2 (%)	NR	NR	NR	
	Placebo (n = 48)	NR	NR	2%	5 (%)	NR	NR	NR	
Li, 2016 [19]	Apatinib (n = 176)	NR	5/181 (2.7%)	Reduction: 21.0%	NR	NR	17/181	NR	
	Placebo (n = 91)	NR	0/92 (0%)	Reduction: 3.3%	NR	NR	3/92	NR	
ATTRACTION-2 Kang, 2017 [17]	Nivolumab (n = 330)	300/330 (91%)	9/330 (3%)	Dose delay 63/330 (19.1%)	NR	5/330 (2%)	33/330 (10%)	34/330 (10%)	
	Placebo (n = 161)	135/161 (84%)	4/161 (2%)	Dose delay 27/161 (16.8%)	NR	2/161 (1%)	8/161 (5.0%)	7/161 (4%)	
JAVELIN Gastric 300 Bang, 2018 [16]	Avelumab + BSC (n = 184)	90 /184[a] (48.9%)	7/184[a] (3.8%)	NR	13/185 (7.0%)	0	NR	17/184[a] (9.2%)	1/184[a,b] (0.5%)
	Chemotherapy + BSC (n = 177)	131/177[a] (74.0%)	9/177[a] (5.1%)	NR	9/186 (4.8%)	1/177[a] (<1%)	NR	56/177[a] (31.6%)	13/177[a,b] (7.3%)
TAGS Shitara, 2018 [18]	Trifluridine/tipiracil + BSC (n = 335)	326/335 (97%)	43/335 (13%)	Delay or reduction: 195/335 (58%)	11/335 (3.3%)	1/335 (<1%)	143/335 (43%)	172/335 (51%)	51/335 (15%)
	Placebo + BSC (n = 168)	157/168 (93%)	28/168 (17%)	Delay or reduction: 37/168 (22%)	2/168 (1.2%)	1/168 (<1%)	70/168 (42%)	64/168 (38%)	14/168 (8%)
CheckMate-032 Janjigian, 2018 [20]	Nivolumab 3 mg/kg (n = 59)	41/59 (69%)[a]	2/59 (3%)[a]	0/59 (0%)	NR	NR	6/59 (10%)[a]	10/59 (17%)	
	Nivolumab 1 mg/kg + ipilimumab 3 mg/kg (n = 49)	41/49 (84%)[a]	10/49 (20%)[a]	0/49 (0%)	NR	NR	21/49 (43%)[a]	23/49 (47%)	

3L+ STUDY Trial Number (Acronym)	Treatment (n)	Overall Rate of Aes, n/N (%)	Discontinuations Due to AEs, n/N (%)	Dose Reductions Due to AEs, n/N (%)	Discontinuations Due to Death, n/N (%)	Treatment-Related Mortality, n/N (%)	Overall Rate of SAEs, n/N (%)	Rate of Adverse Events	
								Grade 3 n/N (%)	Grade 4 n/N (%)
	Nivolumab 3 mg/kg + ipilimumab 1 mg/kg (n = 52)	39/52 (5%)[a]	7/52 (13%)[a]	0/52 (0%)	NR	1/52 (1.9%)[b]	13/52 (25%)[a]	14/52 (27%)	
ATTRACTION-2 (subgroup analysis [Japanese patients]) Kato, 2019 [70]	Nivolumab (n = 152)	129/152 (84.9%)	7/152 (4.6%)	NR	NR	2/152 (1.3%)	35/152 (23%)	24/152 (15.8)	
	Placebo (n = 72)	53/72 (73.6%)	4/72 (5.6%)	NR	NR	1/72 (1.4%)	18/72 (25%)	7/72 (9.7)	
ATTRACTION-2 (subgroup analysis [prior trastuzumab use]) Satoh, 2019 [71]	History of trastuzumab nivolumab (n = 59)	54/59 (91.5%)	3/59 (5.1%)[a]	NR	NR	1/59 (1.7%)	6/59 (10.2%)[a]	NR	NR
	History of trastuzumab placebo (n = 22)	NR	NR	NR	NR	NR	NR	NR	NR
	No history of trastuzumab nivolumab (n = 271)	247/271 (91.1%)	6/271 (2.2%)[a]	NR	NR	8/271 (3.0%)	29/271 (10.7%)[a]	NR	NR
	No history of trastuzumab placebo (n = 139)	NR	NR	NR	NR	NR	NR	NR	NR

AE, adverse event; BSC, best supportive care; FU, fluorouracil; IQR, interquartile range; NR, not reported; SAE, serious adverse event; SoC, standard of care.

[a] All follow-up assessments were completed 24 months after enrollment.

[b] Adverse events leading to death with a causal relation to any study drug.

[c] Median follow-up for overall survival was 7.9 months (IQR, 4.2-13.0).

[d] Median follow-up.

[e] Pulmonary hemorrhage, nose.

[f] Reported as “grade 3 or higher” adverse events.

[g] Reported as “grade 3 to 4” adverse events.

1L Studies	1	2	3	4	5	6	7	Score

Supplemental Table 4. Assessment of risk for bias in the included 1L and post-1L RCTs

The risk for bias assessment used in this systematic literature review was developed by Wagner, et al (2010). The seven criteria are as follows: 1) Was the allocation truly random? 2) Was the treatment allocation concealed? 3) Were the groups similar at baseline regarding the most important prognostic factors? 4) Were the number of withdrawals, dropouts, and losses to follow-up in each group completely described? 5) Was the analysis done by intention to treat? 6) Were the type and schedule of the follow-up similar in the comparison group? 7) Have computed tomography/nuclear magnetic resonance scans been reviewed by an independent radiologist? Responses were either “Yes” (green, +), “No” (orange, -) or “Unclear” (yellow, ?).

Assessment categories for each item were yes (a), no (b), and unclear (c). Based on these criteria, the studies were broadly subdivided into the following four categories of quality:

A) All of the criteria are met: low risk of bias
B) All of criteria 1, 2, and 3 are met; one or more of criteria 4 to 7 are not met
C) Criteria 1 is met, and one or more of criteria 2 to 7 are not met
D) Unclear: insufficient information given

SOS Ryu, 2015 [1]	+	+	+	+	+	+	+	A
AVATAR Shen, 2015 [30]	+	+	+	+	+	+	?	B
Van Cutsem, 2015 [2]	+	+	?	+	+	+	?	C
REAL3 Waddell, 2013 [38]	+	-	+	+	+	+	-	C
Wang, 2016 [35]	+	-	+	+	-	+	-	C
G-SOX trial Yamada, 2015 [22]; Bando, 2016 [23]	+	-	+	+	?	+	+	C
ToGA Bang, 2010 [5]	+	-	+	+	-	+	?	C
Curran, 2009 [29]	+	?	?	?	-	+	?	C
Guimbaud, 2014 [36]	?	?	+	+	+	+	?	D, unclear
TRIO-013/LOGIC Hecht, 2016 [4]	+	?	+	+	-	?	?	C
Kang, 2009 [24]	+	?	+	+	+	+	+	C
Kim, 2014 [31]	?	?	+	+	+	+	?	D, unclear
Qu, 2015 [3]	+	-	+	+	+	+	?	C
EXPAND Lordick, 2013 [32]	+	-	+	+	+	+	+	C
AVAGAST Ohtsu, 2011 [33]	+	?	+	+	+	+	-	C
FLAGS Ajani, 2010 [11]	?	?	+	+	-	+	+	D, unclear
DIGEST Ajani, 2017 [25]	+	-	+	-	+	+	?	C
RILOMET-1 Catenacci, 2017 [8]	+	+	+	+	+	+	?	B
Lu, 2018 [26]	+	-	+	+	+	+	?	C
METGastric Shah, 2017 [9]	+	?	+	+	+	+	?	C
Lu, 2019 [27]	?	-	+	+	+	+	?	D, unclear
Shu, 2017 [28]	+	-	+	+	-	+	?	C
HELOISE Shah, 2017 [6]	+	-	+*	+	+	+	?	C
RAINFALL Fuchs, 2019 [34]	+	+	+	+	+	+	?	B
JACOB Tabernero, 2018 [7]	+	+	+	+	+	+	?	B
JCOG1013 Yamada, 2019 [37]	+	-	+	+	+	+	?	C
2L Studies	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Score**
JapicCTI-090849 Satoh, 2015 [62]	+	-	+	+	-	+	+	C
TyTAN Satoh, 2014 [63]	+	-	+	+	+	+	?	C
Sym, 2013 [49]	?	?	?	+	+	+	?	D, unclear

JACCRO GC-05 Tanabe, 2015 [53]	?	?	+	+	-	?	+	D, unclear
Thuss-Patience, 2011 [15]	+	-	?	+	+	+	?	C
Yi, 2012 [64]	+	-	?	?	-	+	+	C
Study 39 Bang, 2015 [65]	+	?	+	+	+	+	-	C
COUGAR-02 Ford, 2014 [52]	+	-	+	+	+	+	?	C
TCOG GI-0801/BIRIP Higuchi, 2014 [54]	+	?	+	+	-	+	+	C
REGARD Fuchs, 2014 [39]	+	+	+	+	+	+	-	B
RAINBOW Wilke, 2014 [66]; Muro, 2016 [68]; Shitara, 2016 [69]	+	+	+	+	+	+	-	B
WJOG 4007 Hironaka, 2013 [40]	+	-	+	+	-	+	-	C
Kim, 2015 [55]	?	?	+	-	+	+	?	D, unclear
CCOG0701 Nakanishi, 2016 [56]	+	-	+	+	-	+	?	C
JCOG0407 Nishina, 2016 [41]	+	-	+	+	-	+	-	C
Roy, 2013 [12]	?	?	+	+	+	+	?	D, unclear
TRICS Nishikawa, 2015 [57]	+	?	+	+	-	+	?	C
Bang, 2017a [42]	?	-	+	+	+	+	+	D, unclear
GOLD Bang, 2017b [59]	+	+	+	+	+	+	?	B
DREAM Kang, 2018 [48]	?	-	+	-	-	+	?	D, unclear
KEYNOTE-061 Shitara, 2018 [43]	+	-	+	+	+	+	+	C
GATSBY Thuss-Patience, 2017 [44]	+	-	+	+	+	?	?	C
SHINE Van Cutsem, 2017 [45]	?	-	+	+	+	+	?	D, unclear
Lee, 2017 [13]	+	-	-	+	-	+	?	C
KCSG ST10-01 Lee, 2019 [46]	+	-	+	+	+	+	?	C
ABSOLUTE Shitara, 2017 [14]	+	-	+	+	-	+	+	C
Combined 2L and 3L Studies								
Fushida, 2016 [58]	+	-	+	+	-	?	?	C
Kang, 2012 [51]	+	-	+	+	+	+	?	C
Moehler, 2016 [50]	?	+	-	+	+	+	?	D, unclear
Ohtsu, 2013 [60]	+	+	-	+	+	+	?	C

INTEGRATE Pavlakis, 2016 [61]	+	+	+	-	+	+	+	B
Shitara, 2014 [47]	+	-	+	+	-	+	?	C

3L+ Studies	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Score**
Li, 2016 [19]	+	+	?	+	+	-	+	C
Li, 2013 [21]	+	+	+	+	+	+	+	A
JAVELIN Gastric 300 Bang, 2018 [16]	?	-	+	+	+	?	+	D, unclear
ATTRACTION-2 Kang, 2017 [17]	+	+	+	+	+	+	?	B
TAGS Shitara, 2018 [18]	+	+	+	+	+	+	-	B
CheckMate-032 Janjigian, 2018 [20]	?	?	-	+	+	+	+	D, unclear